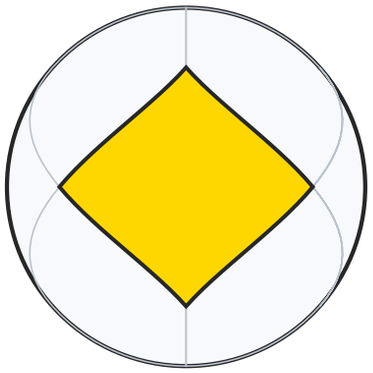
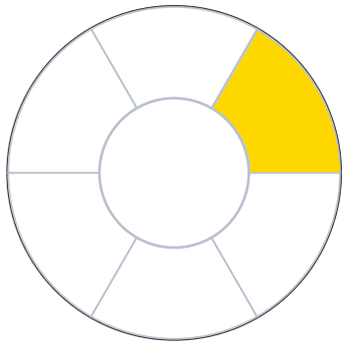
NeST: an exactly-normalized $SO(3)$ density from a Hopf–HEALPix partition tree

A. One equal-volume cell = HEALPix S^2 pixel × Hopf roll fibre S^1 ⇒ each node refines ×8



S^2 base
12 HEALPix pixels

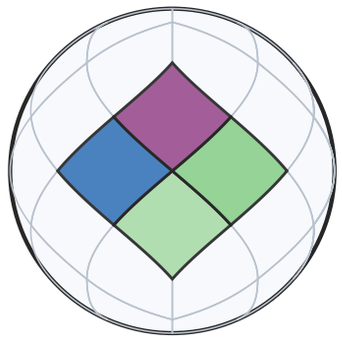
×



roll fibre S^1
6 equal arcs

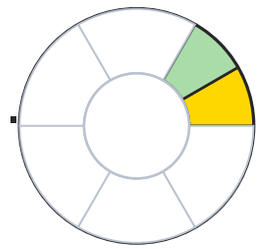
=

one $SO(3)$ cell
equal Haar volume
$N(0) = 12 \times 6 = 72$



pixel ×4
(NESTED)

·

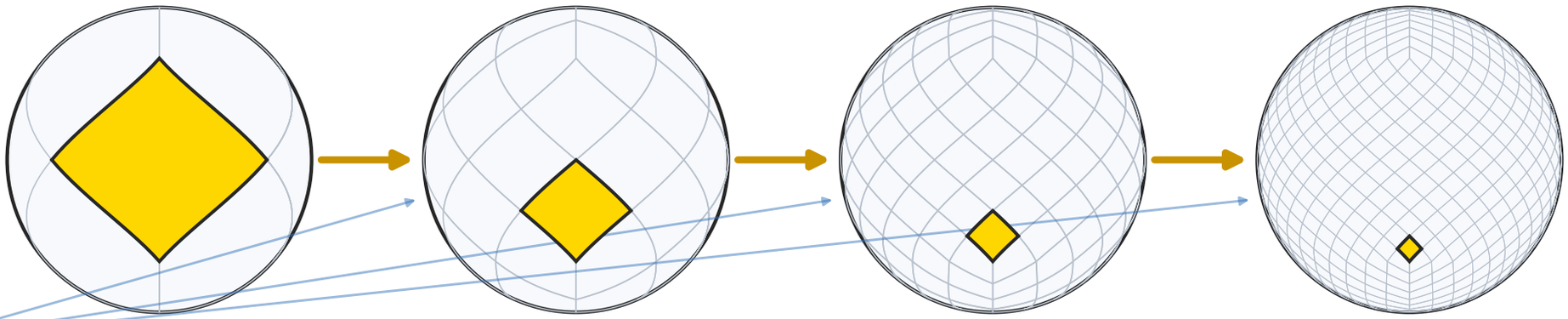


roll ×2

B. The tree: the encoder conditions a per-level 8-way softmax that refines the GT cell

image x

encoder
(ViT-B/16)
→ tokens



$p_0(c_0 \mid x)$ **72-way**

$p(c_1 \mid \cdot)$ **8-way**

$p(c_2 \mid \cdot)$ **8-way**

$p(c_L \mid \cdot)$ **8-way**

gold = the single GT root→leaf path (the cell actually scored); the pixel shrinks ×4 and the roll arc ×2 every level

C.

density = path-product × N/V

$$\log p(R \mid x) = \log p_0(c_0) + \sum_{\ell} \log p(c_\ell \mid \cdot) + \log(N/V)$$

- **Exact normalization** $\int p \, dR = 1$
- **Exact uniform** $\to -\ln \pi^2$
- $O(L)$ **scoring** $L+1 \approx 7$ vs ~ 2.36M